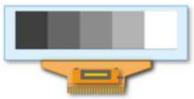
PM-OLED 標準モデル / PM-OLED Standard Models

No.M06857A001
この標準モデルリストは2021年8月時点のもので、予告なく変更させて頂く場合が御座います。
This list is as of August 2021 and subject to change without any advance notice.

開発ステップ Development Phase		Mass Production	Mass Production	Mass Production	Mass Production	Mass Production	Mass Production	Mass Production	Mass Production	Mass Production
モデル名 Model Name		MXS4183	MXS8498	MXS8499	MXS4173	MXS8500	MXS4180	MXS4169	MXS8501	MXS4182
外形図 External View										
ドット数 (W x H) Number of Dots (W x H)	-	60 x 32	96 x 39	128 x 36	128 x 22	128 x 96	128 x 36	96 x 96	192 x 64	128 x 128
ディスプレイサイズ Diagonal Active Area Size	inch	0.5	0.8	0.9	0.9	1.0	1.1	1.1	1.4	1.5
ガラス外形 Glass Size	mm	15.3 x 10.2	22.3 x 11.9	27.6 x 11.6	30.2 x 8.0	25.6 x 22.5	31.9 x 12.8	27.0 x 30.1	40.6 x 19.0	31.6 x 32.9
アクティブエリア Active Area Size	mm	11.25 x 5.99	18.01 x 7.30	22.63 x 6.34	22.63 x 3.874	20.82 x 15.62	26.85 x 7.53	20.13 x 20.13	33.57 x 11.17	26.09 x 26.09
ドットピッチ Dot Pitch	mm	0.188 x 0.188	0.188 x 0.188	0.177 x 0.177	0.177 x 0.177	0.163 x 0.163	0.210 x 0.210	0.210 x 0.210	0.175 x 0.175	0.204 x 0.204
厚み (偏光板付) Total Thickness (with Polarizer)	mm	0.94	1.01	0.99	1.29	1.19	1.04	1.73	1.19	1.49
製品重量 Product Weight	g	0.35	0.68	0.73	0.8	1.7	1.05	2.6	2.2	3.6
インターフェース Interface	-	Serial(SPI,I2C)	Serial(SPI)	Serial(SPI,I2C)	Serial(SPI)	Serial(SPI)	Serial(SPI,I2C)	Parallel	Serial(SPI)	Serial(SPI)
ドライブ系電源電圧 Drive System Power Voltage	V	12.5	13.5	14.0	11.5	14.5	10.7	14.8	13.0	15.0
ロジック系電源電圧 Logic System Power Voltage (LSPV)	V	1.65 to 3.5	2.5 to 3.4	1.65 to 3.5	1.65 to 3.5	1.65 to 3.4	1.65 to 3.5	1.65 to 3.5	1.65 to 3.4	1.65 to 2.1
インターフェース系電源電圧 Interface System Power Voltage	V	1.65 to 3.5	1.65 to 3.4	1.65 to 3.5	1.65 to 3.5	1.65 to 3.4	1.65 to 3.5	1.65 to 3.5	1.65 to 3.4	1.65 to 2.1
カラー Color	-	White	White	White	White	White	White	White	White	White
輝度 (偏光板付) Luminance (with Polarizer)	cd/m^2	450	400	400	400	270	200	140	190	90
色度 CIE (x,y) Color Coordinates	-	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)
階調 Gray Scale	-	2	2	2	2	2	2	16	2	16
使用温度範囲 Operating Temperature	℃	-20 to +70	-20 to +70	-20 to +70	-20 to +70	-20 to +70	-20 to +70	-20 to +70	-20 to +70	-20 to +70
保存温度範囲 Storage Temperature	℃	-30 to +80	-30 to +80	-30 to +80	-30 to +80	-30 to +80	-30 to +80	-30 to +80	-30 to +80	-30 to +80

開発ステップ Development Phase		Mass Production	Mass Production	Mass Production	Mass Production	Mass Production	Under Development
モデル名 Model Name		MXS4046	MXS4064	MXS4035	MXS4145	MXS4039	MXS4184
外形図 External View							
ドット数 (W x H) Number of Dots (W x H)	-	128 x 100	256 x 64	256 x 64	96RGB x 96	160RGB x 59	128 x 64
ディスプレイサイズ Diagonal Active Area Size	inch	1.6	2.9	3.6	1.1	2.1	0.96
ガラス外形 Glass Size	mm	37.0 x 33.7	79.4 x 30.8	99.2 x 30.4	27.0 x 30.1	54.6 x 24.0	26.7 x 19.3
アクティブエリア Active Area Size	mm	31.33 x 24.47	71.39 x 19.04	87.52 x 21.28	20.13 x 20.13	48.93 x 18.03	21.74 x 11.20
ドットピッチ Dot Pitch	mm	0.245 x 0.245	0.279 x 0.298	0.342 x 0.333	0.210 x 0.210	0.306 x 0.306	0.17 x 0.17
厚み (偏光板付) Total Thickness (with Polarizer)	mm	1.53	2.03	2.03	1.73	1.65	1.4
製品重量 Product Weight	g	4.63	11.4	15.2	2.6	4.3	TDB
インターフェース Interface	-	Serial(SPI)	Serial(SPI)	Parallel Serial(SPI)	Parallel	Serial(SPI)	Parallel Serial(SPI,I2C)
ドライブ系電源電圧 Drive System Power Voltage	V	17.0	15.5	16.0	14.8	18.0	12
ロジック系電源電圧 Logic System Power Voltage (LSPV)	V	1.65 to 3.5	1.65 to 3.4	2.4 to 3.5	2.88 to 3.12	2.60 to 3.50	2.8 to 3.3
インターフェース系電源電圧 Interface System Power Voltage	V	1.65 to 3.5	1.65 to 3.4	1.65 to 3.5	1.88 to 3.12	1.60 to 3.50	2.8 to 3.3
カラー Color	-	White	White	White	65k Colors	262k Colors	Yellow(128x16) Blue(128x48)
輝度 (偏光板付) Luminance (with Polarizer)	cd/m^2	100	85	100	80	100	115
色度 CIE (x,y) Color Coordinates	-	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.31 , 0.33)	(0.28 , 0.32)	Blue(0.16,0.27)
階調 Gray Scale	-	16	2	16	65k Colors	262k Colors	2
使用温度範囲 Operating Temperature	℃	-20 to +70	-30 to +75	-20 to +70	-20 to +70	-20 to +70	-20 to +70
保存温度範囲 Storage Temperature	℃	-40 to +80	-40 to +85	-30 to +80	-30 to +80	-30 to +80	-30 to +80

ご注意

1. 標準モデルと仕様は予告なく変更させて頂く場合がございますのでご了承下さい。
2. 仕様は主要箇所のみ記載しております。下記の詳細に関しては営業部門にお問い合わせ下さい。
   ※フレキシブル基板の仕様はお客様のご要望にお応え致します。
   ※単色モデルに関しては各色取揃えております。また、標準モデル以外のカスタムのご要望にもお応え致します。

■お問い合わせをお待ちしております。

【標準品のお問い合わせはこちら】 https://form.jpn.pioneer/topec/oled/standard/sp/
【カスタムのお問い合わせはこちら】 https://form.jpn.pioneer/topec/oled/custom/sp/

東北パイオニア株式会社
外販事業統括部 OLED事業部
〒992-1128 山形県米沢市八幡原四丁目3146-7
phone.**0238-28-7853**
https://jpn.pioneer/topec/oled

Note

1. All models and specifications in this lineup are subject to change without any advance notice.
2. The specifications in this lineup show essential information only. Please contact our sales department for the following details.
   ※Specifications of Flexible Printed Circuits(FPC) are available to meet customers'requirements.
   ※Mono-color models have a wide variety of color options. Also custom models are available by customers' requirements in addition to the models shown in this lineup.

■Looking forward to your inquiry.

【For Sales Inquiry】 https://form.jpn.pioneer/topec/oled/sales/sp/
【For Technical Inquiry 】 https://form.jpn.pioneer/topec/oled/tech/sp/

TOHOKU PIONEER CORPORATION
OLED Business Unit
External Sales Business Management
4-3146-7 Hachimanpara, Yonezawa,
Yamagata 992-1128, JAPAN
phone.**+81-238-28-7853**
https://global.pioneer/topec/oled

Semi-customizable Standard Product Lineup

Realizes customization according to your application at low introductory costs

Color Variations

Customization of display colors



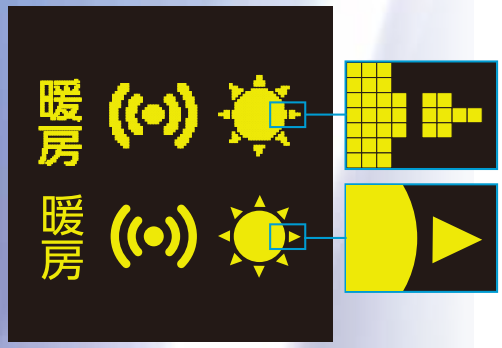
Based on our standard product lineup (white monocolor), display colors (emission colors) can be changed. Color changes never sacrifice mountability and display content because the external form, display size, and resolution are identical to base models.
We offer a selection of 7 display colors in total, taking advantage of OLED-specific vivid colors. You can choose a display color that matches the design and application of your equipment.

Color	Feature
WHITE	Base color matches a wide range of equipment and applications with elegant colors.
BLUE※	Dark blue coloration gives a unique atmosphere to the display.
SKY BLUE	Clear blue-based color, excellent in visibility has a near-future sense in color tone.
GREEN	Vivid colors and excellent contrast.
YELLOW	Light emitting luminance easily increased for excellent colors and visibility.
ORANGE(AMBER)※	Warm colors that reduce glare.
RED	Vivid colors and excellent readability, suitable for warning displays, etc.

※The display luminance is lower compared with base models.

Segment Display

Customization of displays that faithfully reflect designs



While the full-dot displays of our standard product lineup are adaptable to characters, graphics, and other universal displays, it is considered that sufficient resolution and legibility may not be obtained depending on your application such as cases where reproduction of icons and other designs is important, and in the case of complicated Kanji displays.
We can customize our standard product lineup to character type (fixed icon type) displays without changing their external forms and display sizes.
By creating icon-shaped light emitting pixels, it is possible to achieve sharp representations that faithfully reflect original designs. By dividing ON/OFF motion units within one icon, minimum dynamic representations such as a battery indicator and reception strength indication are possible.
We offer high-visibility displays making full use of high contrast, a main feature of OLED displays.

Multicolor Design

Customization of displays that enables “effect color”

[Design examples by the mask-shadowing vapor deposition patterning method]



Shadow mask patterning design example

Shadow mask patterning display example

By dividing the display area of our standard product lineup (white monocolor) and placing multicolored OLED elements, displays that are partly different in display color* can be arranged. This realizes greater expressive display representations, including proper use of a main display and an alert display and the effect color of the icon portion.
This is applicable not only for full-dot type displays but also character type displays.

※There are restrictions on the combination of emission colors.

[Design examples by our unique multicolor method]



Full-dot model

Segment model

Multi-coloring by conventional mask-shadowing vapor deposition patterning method enables multicolor designs at high definition, but restricts the division of the display area. As the degree of freedom of the multicolor display area is considerably improved by our unique multi-coloring method, we can offer designs according to your request.

Higher Luminance

Customization of displays that pursue higher brightness



In principle, dot-matrix type PMOLED displays comprised of duty-driving self-light-emitting elements are characterized by the fact that the maximum display luminance (brightness) decreases as the display area and the number of dots increase.
For our standard product lineup, white elements with the highest level of efficiency in the industry are adopted as mass-produced products, and some models can also be arranged to specifications with the display luminance further improved by our unique technology (special panel design and drive/control technology) in combination.
We can offer displays suitable for applications requiring high luminance, in such cases where outdoor use or use in other bright places is assumed, and for the blackout designs that minimize “display frames” when not using the display.

TOHOKU PIONEER CORPORATION OLED Business Unit
External Sales Business Management
4-3146-7 Hachimanpara, Yonezawa City, Yamagata 992-1128, Japan
phone. **+81-238-28-7853** https://**global.pioneer/topec/oled**

TRANSPARENT

See-through Future Display

Transparent OLED

Our exquisite vapor deposition technology and optical design technology realize excellent transparency.
A wide range of information can be superimposed on display without losing background visibility.

■ Vibrant Self-light-emitting Display and Excellent Transparency

The vivid display of OLED, a self-light-emitting element, demonstrates its true value even with transparent displays using the background as a watermark.
There is no background coloring or dark contrast degradation by backlight. Also, for transmission images during display-off conditions, the appearance of segments is completely eliminated by optimizing the optical design.

Our OLED
shadow mask patterning
+
optical design



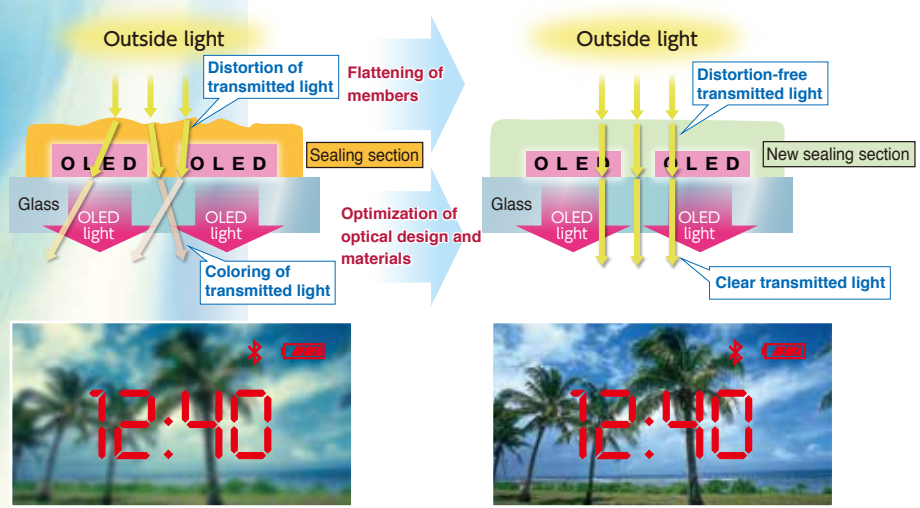
Clear-cut display



Patterns are invisible at non-display time.

■ Compatibility between Space-saving Structure and Transmission Image Performance

Slimness and narrow bezel-oriented high-specification transparent OLED displays tend to impact integrity, including distortion of transmission images, due to structural restrictions. Our products have improved device flatness by optimization of materials and processes, achieving high dimensional compatibility for both space saving and transmission image performance.



■ We Offer Transparent OLEDs Tailored to Your Need

Through integration with our unique technology, it is possible to customize transparent OLEDs according to your individual casing and information display.
We contribute to improving your product value through changes in panel shape, OLED-specific multicolor display, etc.

	Transmittance	Additional options: Special substrate	Multicolor	Free-form	Narrow bezel	Thin curve	Application examples
High-end model	90%	●	●	●	●	●	HUD, AR smart glasses
Middle-range model	80%	●	●	●	●		Binoculars, Viewfinders
Standard model	70%	●	●				Laser distance meters, Wristwatches

In addition to the above, we make it our business to meet various customer requirements.

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ULTRA-THIN

Pioneer

To the Ultimate in “Thinness”

Ultra-thin OLED

Our unique OLED sealing and back protection structures realize outstanding thinness.
The thinnest display in the PM-OLED industry enables information display in limited spaces.

■ Space-saving Thin Display with One-Piece Members

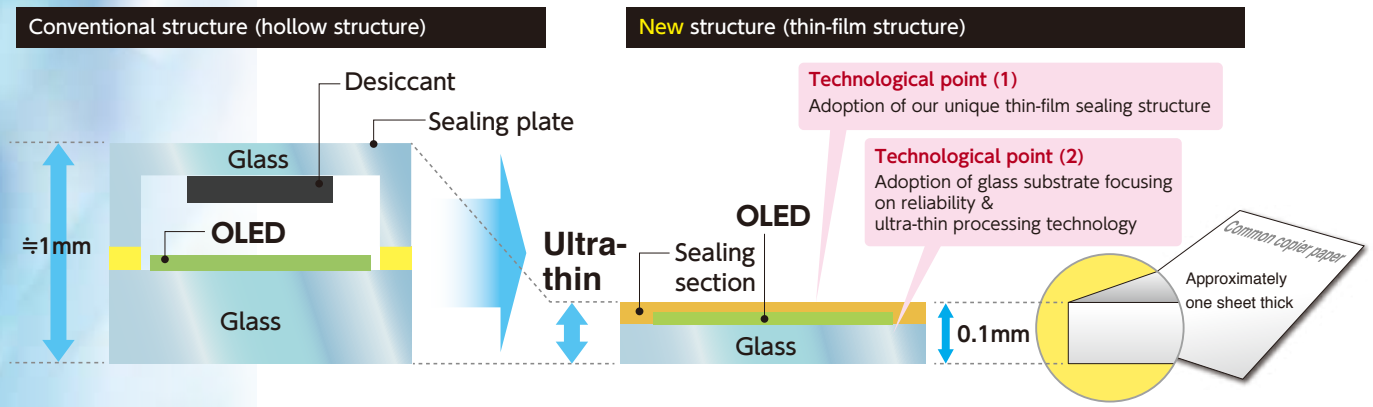
We eliminate the back-sealing plate by combining our unique thin-film encapsulation and protection structures and also unify primary members constituting thickness to achieve a thickness of 0.1mm, which is the thinnest in the PM-OLED industry.

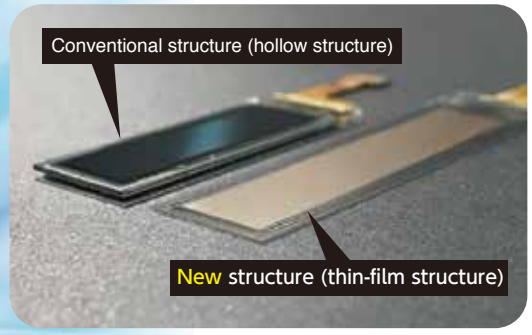
■ Realization of High-reliability Display that Combines Thinness and Toughness

Through the adoption of glass as a base material, we ensure a reliable water vapor barrier property just like regular type OLEDs. It can also be used even in high temperature environments where film-based displays cannot be used.

■ Glass-based Bendable Display Realized by our Unique Process

Even glass can be bent by thinning it as much as possible. Our unique process ensures a bending radius that is necessary and sufficient on miniature displays, an advantage of PMOLEDs.





Ultra-thin OLED application examples

Example 1
Card-type terminal

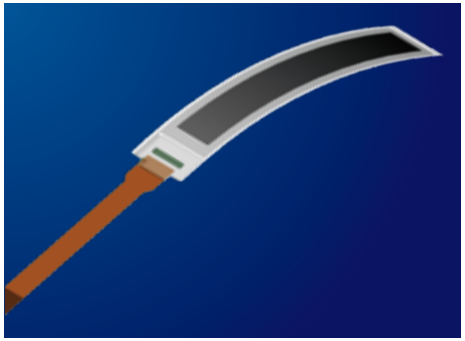


Can be mounted on ultra-thin casings

Example 2
Wearable device



Realizes a display that fits a curved surface





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External Sales Business Management
4-3146-7 Hachimanpara, Yonezawa City, Yamagata 992-1128, Japan
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HMI INTEGRATION

View, Touch, and Manipulate.

In-cell Type Touch Panel OLED

Through the adoption of the special OLED driver IC, the touch sensor function is incorporated in the OLED panel.
For panels only for “viewing,” “touchable and manipulatable” functions are added with space-saving, parts-saving & simple mechanism.

■ Realization of Space Saving by Reducing the Number of Parts

The in-cell type touch panel OLED eliminates the need for the conventionally required touch panel film, dedicated IC, FPC, etc. It not only reduces the number of members but also contributes to saving space.

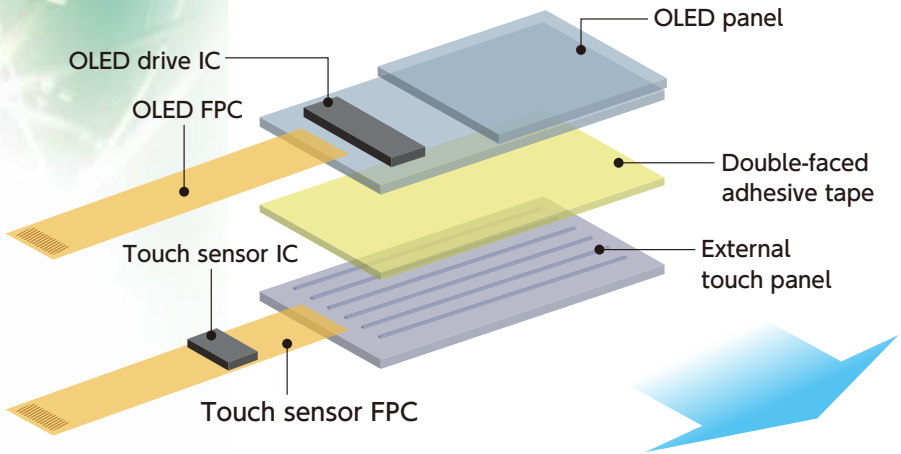
■ Simplification of Interface, Facilitating Electrical Connections

For the in-cell type touch panel OLED, the drive circuits that would be separately required for the OLED panel and touch panel can be integrated into one. Smart connection to your products is realized.

■ Addition of the Touch Function without Impairing the Display Integrity

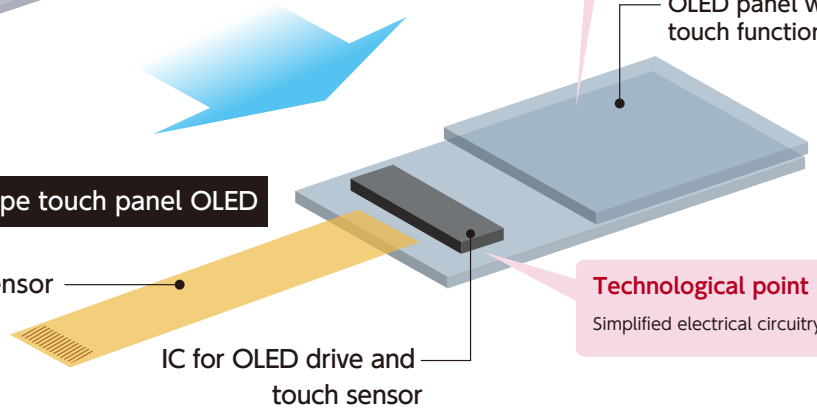
The conventional structure in which a touch panel is affixed on the panel surface has optical changes, such as distortion of the display and changes in color due to wiring of the touch panel, double-faced adhesive tape, etc. The in-cell type structure with a built-in touch panel can provide a panel that is free from such optical changes.

[Conventional structure] Out-cell type touch panel OLED



Technological point (1)
Thin and lightweight due to the reduced number of component parts

[New structure] In-cell type touch panel OLED



Technological point (2)
Simplified electrical circuitry

■ Easily Combined with Our Space-Saving Technology and Applicable in Various Fields

Combination with our space-saving technology

Application example 1



Bluetooth speaker

An OLED panel with a touch function is installed along the curved surface of the casing.
Fast-forward and other intuitive operations can be performed without impairing design features.

Combination with our space-saving technology

Application example 2



Credit card with an electronic payment option

The information display with a touch function can be made without changing the card thickness. The electronic money balance, etc., on the card can be displayed.

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NARROW BEZEL

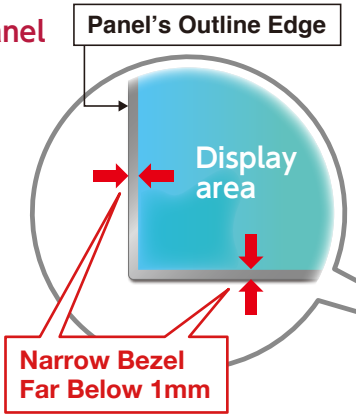
FREE-FORM

Four-sided Ultra-narrow Bezel OLED

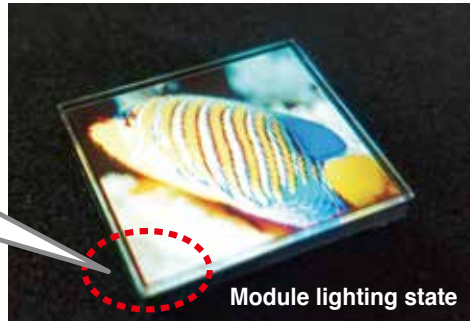
New value of PM-OLEO brought about by higher-precision processing and innovation in drive circuit mounting technology

Four-sided ultra-narrow bezel panel structure enables flexible layout

Through the adoption of high-precision members and the sophistication of panel production technology, the seal area and wiring area are narrowed down to the extent possible. The RGB full-color display fully extends to the outer edges of the panel to give a sophisticated impression that surpasses common knowledge of the conventional PM-OLED.
In addition to enabling display placement in a narrow space within a customer's products, our OLED contributes to improving product designability and functionality by using multiple display devices to create surface height differences.



Realizes the ultra-narrow bezel on all four outer circumference sides!





Surpasses the Boundaries of Common Knowledge

A shift from "fitting the casing design to box-shaped displays" to "displays fitting the casing design"

Free-form OLED



■ Chamfered design



■ Hexagonal design



■ Other designs

- Circular design
- Oval design
- Triangular design
- Rhombic design

Combination of Free-form Glass Cutting Technology and Flexible Module Design

With the diversification of products equipped with small displays, the need for specially shaped displays, including chamfered and circular configurations, continues to grow.
Possessing the most advanced glass cutting equipment that is compatible not only with rectangular shapes, and in combination with custom display module design technology that we have cultivated over many years, we can also propose special shape displays according to customer requirements. Free-form is available with transparent OLED as well as the regular type OLED. We realize a display that maximizes the utilization of your casing design

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External Sales Business Management
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SPECIAL LIGHT SOURCES

Pioneer

The Appeals of Organic EL Devices not Limited to Just Information Display

Customizable Special Light Sources

Combining the advantages of surface emitting devices with optical technology to realize thin and lightweight light source units for special applications.
In addition to display devices, we offer OLED panels that meet your diverse needs.

■ Features of OLED panels

Thin
Because they are thin and bendable in layout, we can propose space-saving light sources that meet your needs.

Surface-emitting
For OLEDs arrayed on one glass plate, light sources can be arranged within a plane in free form in combination with high-definition patterns.

Directivity
Light-emitting directivity can be changed according to light source applications.

Free-form
Not only square shapes, but also round shapes, triangular shapes, and various other designs are possible.

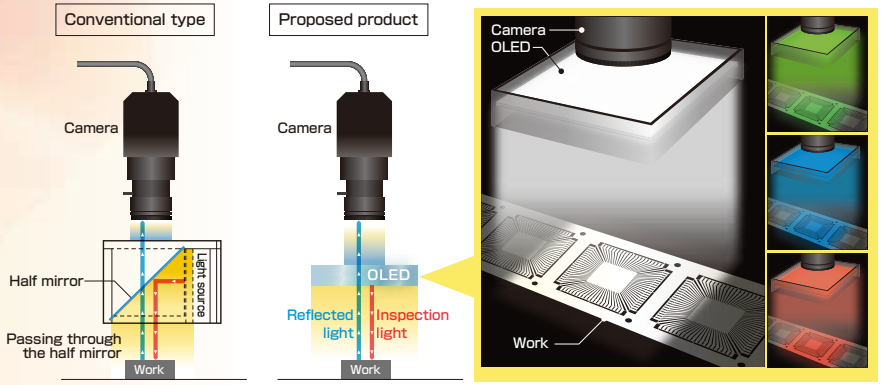
Emission colors
Colorful emission adjustments are possible according to need from light's three primary colors through white light emission high in color rendering properties.

Transparent
By making the light-emitting area transparent, this enables superimposing on the background display. In addition, we can propose single-sided and double-sided light emissions, etc., according to your application.

■ Application Examples of Usage of OLED other than Display Devices

Light source for inspection

Proposal of a coaxial multicolor light source realizing high uniformity and space saving

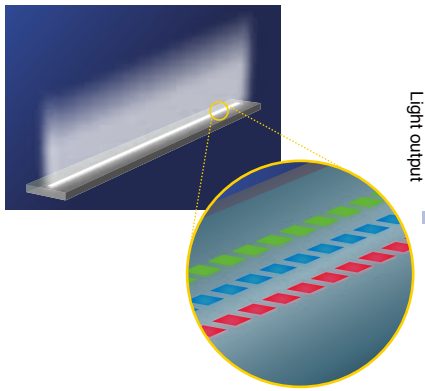


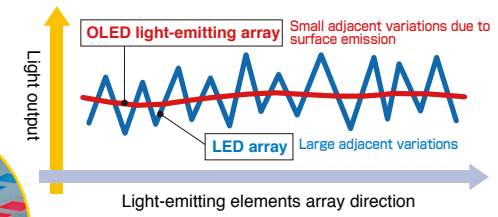
Utilizing the characteristics of surface light sources, we can provide thin and high-luminance light sources suitable for robot-embedded systems. In addition, it is possible to provide highly uniform irradiation and emission color changes at close range, which was difficult with conventional coaxial lighting, making it possible to provide inspection environments according to each workpiece.

Thin | Surface-emitting | Emission colors | Transparent

Linear light-emitting array

Potential as a special light source brought about by monolithic integration of OLEDs and resonators





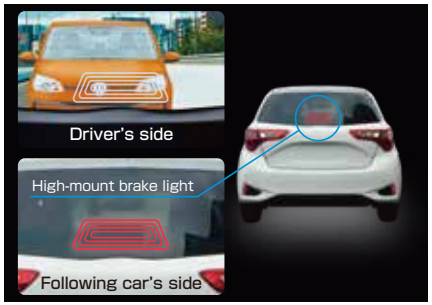
Multiple uniform and high-power light sources utilizing the characteristics of OLEDs are arranged within a plane. They can be expanded to array-form light-emitting devices and other applications requiring high visibility.

Surface-emitting | Emission colors | Directivity

High-mounted brake light

Lamp application proposals, such as beacon lights that do not hinder rear view confirmation





When the OLED panel is used as a high-mount brake light, the rear-view area, which was hidden by conventional brake lights, can be widely secured, while the visibility of the following car is ensured. In addition, the single-sided emission, which was difficult to realize with conventional transparent devices does not interfere with rear view vision even when the lamp is turned on.

Surface-emitting | Free-form | Transparent

TOHOKU PIONEER CORPORATION OLED Business Unit
External Sales Business Management
4-3146-7 Hachimanpara, Yonezawa City, Yamagata 992-1128, Japan
phone. **+81-238-28-7853** https://**global.pioneer/topec/oled**

History and Achievements of Tohoku Pioneer's Organic EL Business

Pioneer

Total of OLED panel shipments:
168,000,000 or more
(as of the end of August 2019)







Mass production of transparent multicolor OLEDs started — 2019

Mass production of ultra-thin (bendable) OLED displays started — 2017

Mass production of ultra-narrow bezel OLED displays started — 2016

Geo-Cosmos at Miraikan
(National Museum of Emerging Science and Innovation)



Full-color OLED displays for large video equipment — 2014



Vehicle-mounted meter display (monocolor) for company B — 2010



Vehicle-mounted meter display (monocolor) for company A — 2008

World's first

Mass production of full-color OLED displays using phosphorescent materials started — 2004



Cell-phone sub-displays (area colors in Japan)



Mass production of full-color OLED displays started — 2003

Japan's first



2002

2000 — Cell-phone main displays (area colors in North America)

Pioneer CD receiver DEH-P9000



1999 — Supply of OLED displays (area colors) for car stereos started

1997 — **Mass production of monocolor OLED displays started**



1995 — Development for mass production started

World's first

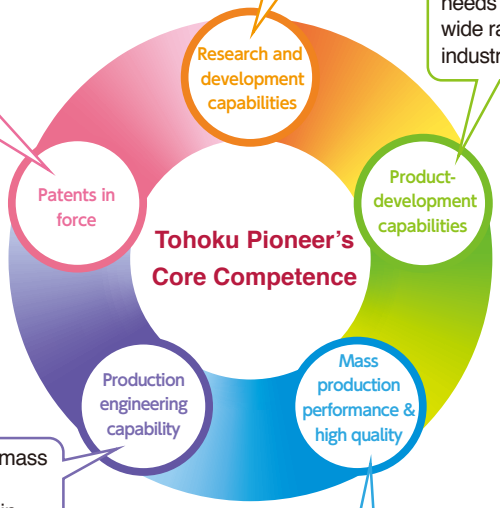


Pioneer FM multiple receiver GD-F1

Number of patents registered: 588 (Pioneer & Tohoku Pioneer/including overseas applications)
※ As of August 2019

New value creation of organic ELs under the Pioneer Group's consistent research and development system since the world's first mass production success

Development of high-value added products according to various customer needs across a wide range of industries



Realization of mass production of transparent, thin, narrow-bezel, multicolor and other products

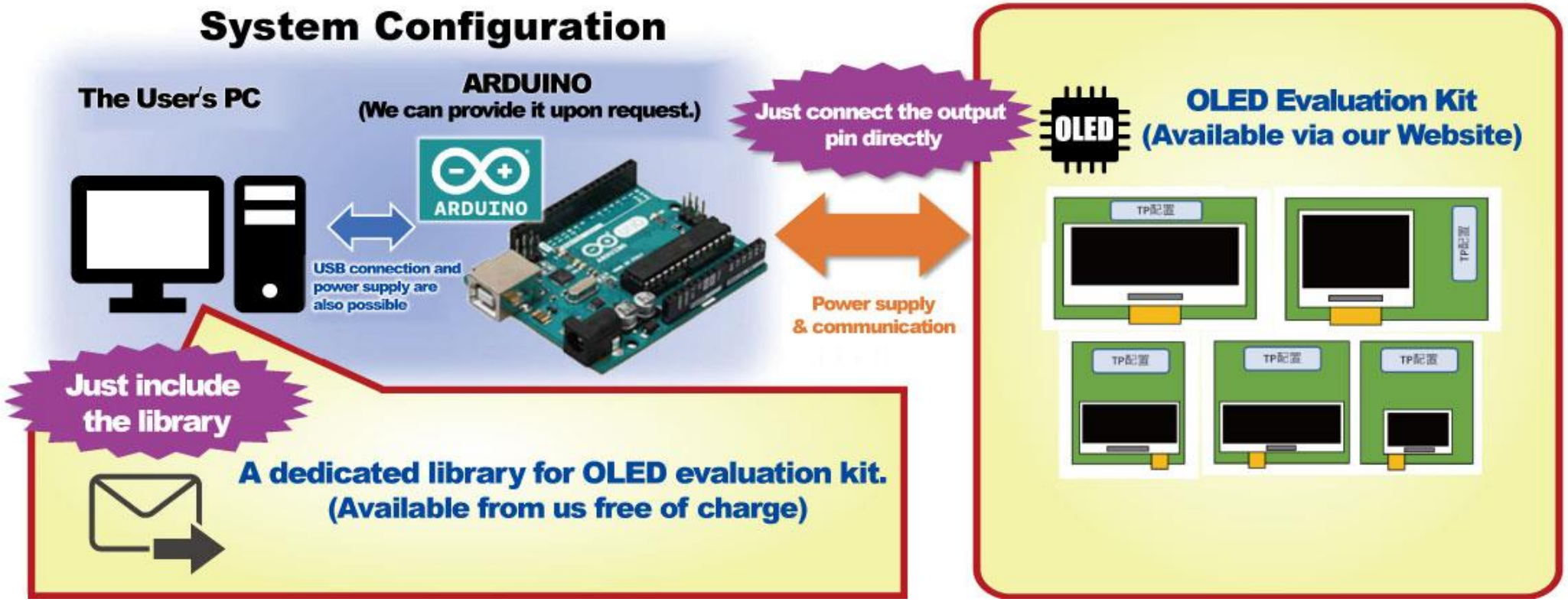
High-quality mass production exceeding 168,000,000 in total
(as of the end of August 2019)

By combining these advantages, we will continue to offer exact solutions that enhance convenience and functionality of your products as a group of specialists in organic EL display technology.

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OLED Module Evaluation Kit

Offering **OLED Module Evaluation Kit** that can be controlled by **ARDUINO**
OLED module simple evaluation environment using a handy development solution



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phone. +81-238-28-7853 https://global.pioneer/topec/oled